

PUDINO, Russian Federation

WMO: 293130

Lat: 57.5627N

Lon: 79.4289E

Elev: 95

StdP: 100.18

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-39.3	-35.9	-42.1	0.1	-39.1	-38.7	0.1	-35.7	7.7	-13.2	7.0	-10.9	0.7	30	0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	28.9	18.7	27.2	18.0	25.4	17.4	20.6	25.7	19.5	24.6	18.5	23.5	3.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.8	13.8	23.1	17.6	12.8	21.8	16.6	12.0	20.8	59.9	25.7	56.2	24.6	53.0	23.5	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.7	6.6	5.9	-42.3	31.9	3.6	1.8	-44.9	33.2	-47.1	34.2	-49.1	35.2	-51.8	36.5		
(5)			-42.1	22.5	3.1	1.5	-44.3	23.6	-46.1	24.4	-47.8	25.3	-50.0	26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.1	-19.9	-15.7	-6.7	1.7	9.1	15.8	17.7	14.5	8.6	1.3	-9.8	-16.5
	DBStd	14.39	8.58	7.73	6.37	5.87	5.67	4.49	3.44	3.68	4.42	5.17	8.84	9.29
	HDD10.0	4286	926	721	518	254	88	8	0	5	78	272	594	821
	HDD18.3	6729	1184	954	777	500	289	98	54	127	294	528	844	1079
	CDD10.0	670	0	0	0	4	62	182	239	146	35	2	0	0
	CDD18.3	71	0	0	0	0	4	22	34	9	1	0	0	0
	CDH23.3	884	0	0	0	1	95	307	363	108	11	0	0	0
	CDH26.7	195	0	0	0	0	26	74	78	16	1	0	0	0
Wind	WSAvg	2.7	2.5	2.7	3.2	3.3	3.1	2.4	2.1	2.2	2.5	2.8	2.9	2.8
Precipitation	PrecAvg	525	25	19	22	25	45	68	79	77	51	42	40	33
	PrecMax	713	55	36	56	58	116	171	187	145	84	98	66	69
	PrecMin	433	2	1	1	6	7	19	9	12	5	19	10	8
	PrecStd	70	14	7	15	12	25	33	38	34	20	19	13	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.3	1.2	9.2	19.7	29.9	30.7	31.2	28.7	24.7	17.0	5.7	0.7
		MCWB	-1.9	-0.2	3.6	11.7	16.8	18.6	20.6	19.3	15.7	10.1	4.0	0.0
	2%	DB	-3.0	-0.8	5.7	16.2	26.0	28.9	28.7	26.3	21.7	13.7	3.9	-0.3
		MCWB	-3.7	-1.9	1.2	9.0	14.8	18.1	19.4	18.1	14.5	8.8	2.7	-1.0
	5%	DB	-5.1	-2.4	3.6	13.6	22.8	26.9	27.1	24.3	18.8	11.0	1.9	-2.3
		MCWB	-5.7	-3.5	0.4	7.6	13.1	17.5	18.7	17.5	13.0	7.5	1.1	-2.9
	10%	DB	-7.2	-4.6	1.8	10.7	19.9	24.8	25.4	22.0	16.4	8.7	0.5	-4.4
		MCWB	-7.7	-5.7	-0.4	5.8	12.1	17.0	18.1	16.8	11.9	6.1	-0.1	-5.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.8	-0.1	3.9	12.3	17.4	21.2	22.5	21.0	17.0	11.3	4.6	0.3
		MCDB	-1.4	0.8	8.6	17.8	27.8	27.1	28.5	25.7	22.3	15.2	5.2	0.6
	2%	WB	-3.6	-1.7	2.0	9.9	15.6	19.8	21.0	19.6	15.2	9.6	2.8	-1.0
		MCDB	-3.1	-0.8	4.4	15.1	23.6	25.6	26.1	24.3	20.2	12.7	3.8	-0.4
	5%	WB	-5.7	-3.4	0.8	7.9	14.1	18.7	19.9	18.4	13.7	7.9	1.1	-2.9
		MCDB	-5.1	-2.5	2.9	12.8	21.3	24.7	24.9	22.4	18.3	10.6	1.7	-2.3
	10%	WB	-7.7	-5.7	-0.3	6.0	12.5	17.7	19.0	17.1	12.3	6.1	0.0	-5.0
		MCDB	-7.2	-4.6	1.8	10.6	19.1	23.2	23.7	21.1	15.8	8.4	0.6	-4.4
Mean Daily Temperature Range	5% DB	MDBR	9.4	11.1	11.3	10.7	11.9	11.9	11.1	10.3	9.8	7.1	7.1	8.2
		MCDBR	6.8	7.8	12.5	14.9	17.4	15.8	14.4	14.6	14.8	11.4	5.3	7.0
		MCWBR	6.7	7.2	8.9	8.7	7.9	7.0	6.6	7.6	8.3	7.3	5.0	6.9
	5% WB	MCDBR	6.9	7.2	9.8	13.8	15.4	13.6	12.5	11.8	13.1	10.7	5.0	6.9
		MCWBR	6.8	6.9	7.4	8.5	7.9	6.9	6.4	6.8	7.8	7.3	4.8	6.9
Clear-Sky Solar Irradiance	taub	0.279	0.312	0.366	0.397	0.399	0.407	0.443	0.412	0.366	0.346	0.307	0.279	
	taud	2.014	2.009	1.939	2.087	2.256	2.314	2.198	2.272	2.358	2.253	2.075	1.961	
	Ebn at Noon	538	692	751	804	834	833	787	787	772	663	496	402	
	Edh at Noon	59	94	134	136	123	118	130	113	89	75	56	47	
All-Sky Solar Radiation	RadAvg	0.69	1.68	3.15	4.39	5.01	5.70	5.14	3.95	2.65	1.29	0.58	0.40	
	RadStd	0.08	0.13	0.21	0.42	0.50	0.60	0.44	0.43	0.32	0.12	0.11	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page